

Smart Car Redesign & Autonomous Navigation

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Project Objectives



REDESIGN

Redesign the Freenove Smart Car for Improved Efficiency & Reliability



OPTIMIZE

Reduce Power Consumption & Simplify Wiring



VISION

Implement Vision-based Navigation & Ultrasonic Obstacle Detection



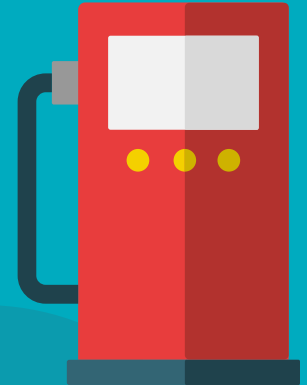
NAVIGATION

Integrate AprilTag Detection for Autonomous Navigation



MANUFACTURE

Develop a Stable & Manufacturable Mechanical Structure



High-Level Overview

Software
Development

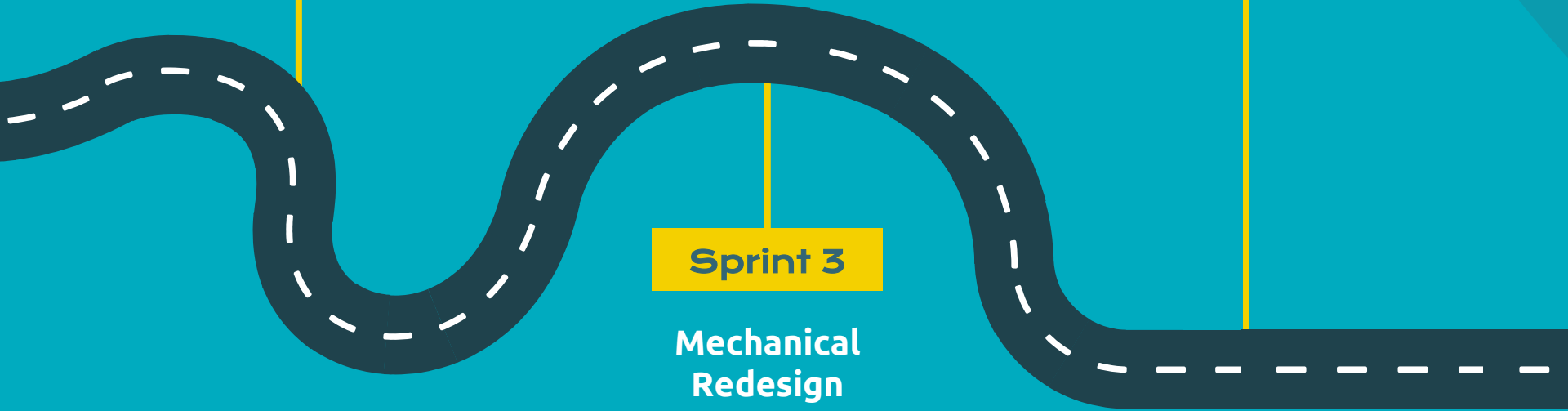
Sprint 2

System Integration

Showcase

Sprint 3

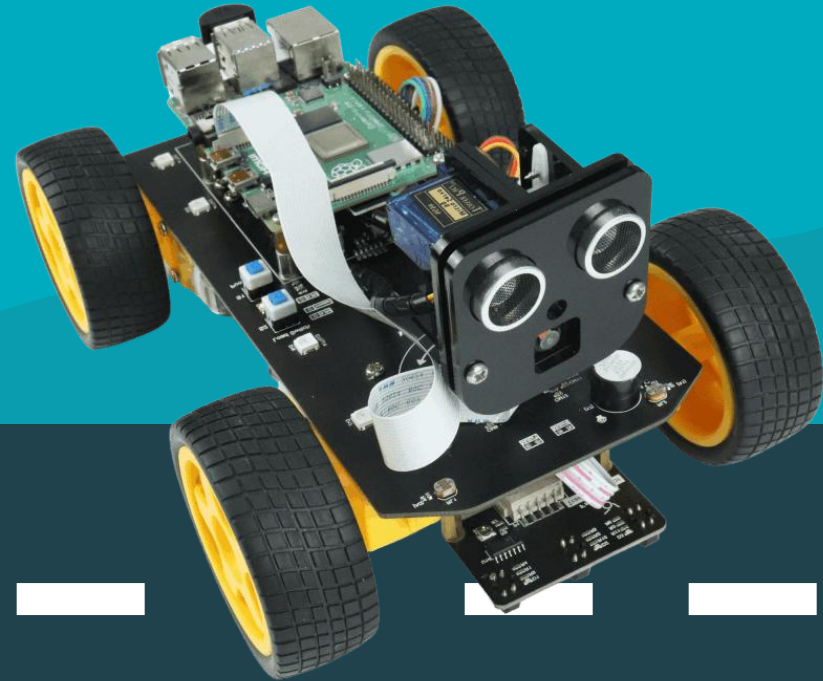
Mechanical
Redesign



Redesign Comparision

4WD Smart Car

Our Solution



Comprehensive Breakdown

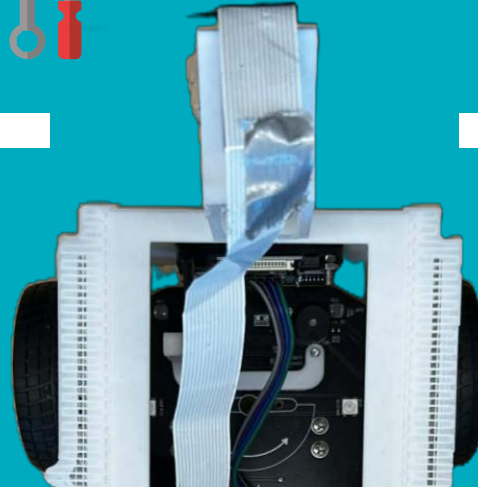
Mechanical Redesign

- Converted 4WD → Front-Wheel Tank Drive
- Rear Ball Mount for Stability
- Custom Durable & Lightweight Chassis
- Fixed Camera Mount
- Reduced Wiring & Component Count



Software & Vision

- Grayscale Image Processing
- Obstacle Detection with Ultrasonic
- AprilTag Detection with OpenCV
- State-Based Motion Control
- Tank Drive Motor Control Logic

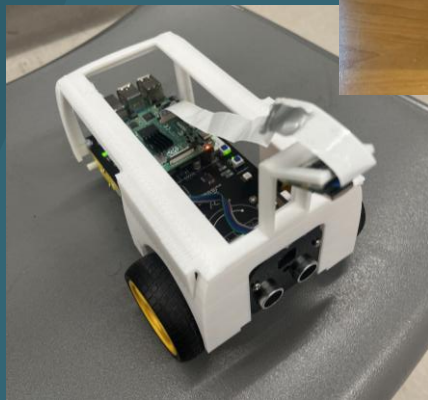
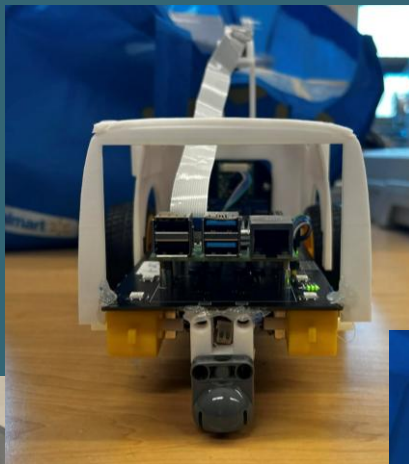




LIVE DEMO







The background is a stylized illustration of a car's interior from the driver's perspective. It features a dark steering wheel on the left, a dashboard with several circular gauges and buttons in the center, and a rearview mirror at the top. The background is a solid light blue color.

QUESTIONS & COMMENTS?

MSE-02